Truth Lemma for the quotient term model (with identity)

The Open Logic Project contributors

openlogic:first-order-logic/completeness:local-snapshot-20261013;
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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-first-order-logic-completeness/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the author's phrase 'just as in the proof of' cites the no-identity Truth Lemma at source line 1312; every case other than identity is unchanged between the two models

Editorial note. The label lem:truth names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lem environment in the unit source.

Allowed standard prerequisites (editorial).

- Truth Lemma for the term model. Same unit, \ollabel{lem:truth} (registered card, source line 1312). The induction skeleton for the non-identity cases carries over to the quotient model unchanged.

Lemma 0.1. $[M]{\approx} \vDash A$ *iff* $A \in \Gamma^*$ *for all sentences* A.

Proof. By induction on A, just as in the proof of 0.1. The only case that needs additional attention is when $A \equiv t = t'$.

$$\begin{aligned}[M]{\approx} \vDash t = t' &\text{ iff } [t]{\approx} = [t']{\approx} \text{ (by definition of } [M]{\approx}) \\ &\text{iff } t \approx t' \text{ (by definition of } [t]{\approx}) \\ &\text{iff } t = t' \in \Gamma^* \text{ (by definition of } \approx).\end{aligned}$$

□